

# Curriculum Vitae

**Name:** David Andrew Snyder  
**Position:** Postdoctoral Research Scholar  
Electrical and Systems Engineering  
University of Pennsylvania  
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[[Google Scholar](#)]  
[[Personal Website](#)]  
[[Github](#)]

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## Academic Positions

Sep 2025 – Present **Postdoctoral Research Scholar, Electrical and Systems Engineering**  
*University of Pennsylvania*, Philadelphia, PA.  
Advisor: Nikolai Matni and George Pappas

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## Education

Sep 2019 – Aug 2025 **Ph.D., Mechanical and Aerospace Engineering**  
*Princeton University*, Princeton, NJ.  
Advisor: Anirudha Majumdar

Aug 2014 – Dec 2018 **B.S., Aeronautical Engineering**, summa cum laude  
**B.S., Economics**, summa cum laude  
*University of Maryland*, College Park, MD.

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## Honors and Awards

2025-present **Penn AI Fellowship**, *University of Pennsylvania*.  
One-year (renewable) fellowship to support postdoc scholarship and travel related to research in machine learning, artificial intelligence, and robotics.

2025-2026 **Academic Grant**, *Toyota Research Institute*.  
Half-year (six-month) academic grant supporting a postdoctoral researcher position at the University of Pennsylvania.

2022 **MAE Crocco Award for Teaching Excellence**, *Princeton University*.  
Award presented for excellence in teaching in the Department of Mechanical and Aerospace Engineering at Princeton University. This award was presented for MAE 345/549: Introduction to Robotics.

2019-2024 **Gordon Y.S. Wu Fellowship in Engineering**, *Princeton University*.  
Five-year fellowship that provides one year of full funding and four years of stipend supplement for supporting graduate study.

2019-2024 **NSF Graduate Research Fellowship (GRFP)**, *National Science Foundation (NSF)*.  
Five-year fellowship (three years of active support) for supporting graduate study.

2014-2018 **Benjamin Banneker Scholarship**, *University of Maryland*.  
Four-year full scholarship to the University of Maryland.

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## Invited Talks

May 2026 [IT5] *Physics-Informed Learning of Feedback-Linearizing Representations*  
Northeast Systems and Control Workshop, Princeton University, Princeton, NJ.

Mar 2026 [IT4] *Towards a Unified Framework for Rigorous Robotic Evaluation*  
RGSO-CARS Seminar, University of Delaware, Newark, DE.

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|--------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apr<br>2025  | [IT3] | <i>The Robot Evaluator Problem: How Do We Know Good Policies When We See Them?</i><br>CPS Seminar, University of Southern California, Los Angeles, CA.      |
| Mar<br>2025  | [IT2] | <i>The Robot Evaluator Problem: How Do We Know Good Policies When We See Them?</i><br>AGH Group Seminar, University of Pennsylvania, Philadelphia, PA.      |
| July<br>2022 | [IT1] | <i>Nonlinear Control, Lyapunov Theory, and Applications to Online Learning</i><br>Google AI Lab Research Presentation Series, Google AI Lab, Princeton, NJ. |

## Industry Experience

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|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2024 –<br>Sep 2024 | <b>Graduate Research Intern</b><br><i>Toyota Research Institute</i> , Trustworthy Learning under Uncertainty Team<br>Researched sample-efficient evaluation procedures for hardware evaluation, as applied to the Large Behavior Model suite developed at TRI. This work formed the basis of STEP [CP8], presented at RSS 2025. |
| Sept 2021 –<br>Aug 2022 | <b>Graduate Research Intern</b><br><i>Google AI Laboratory</i> , Advisor: Elad Hazan<br>Researched online learning methods for high-level robot decision making. This work formed the basis of OLC [CP6], presented at CoRL 2023.                                                                                               |
| Jan 2019 –<br>Aug 2019  | <b>Technical Aide</b><br><i>JHU Applied Physics Laboratory</i> , FP Sector<br>Continuation of undergraduate work focusing on numerical control design and evaluation for nonlinear systems.                                                                                                                                     |

## Teaching

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| Fall ‘21        | <b>Assistant in Instruction, MAE 345/549: Introduction to Robotics</b><br><i>Taught by A. Majumdar at Princeton University.</i><br>An introduction to the basic theoretical and algorithmic principles behind robotic systems, with project-based assignments on quadrotors. Topics include motion planning, control, localization, mapping, and vision. Coursework incorporates Python-based numerical problems via structured Jupyter Notebooks.                                                                                                                                                |
| Spring ‘21, ‘22 | <b>Assistant in Instruction, MAE 206: Introduction to Robotics</b><br><i>Taught by A. Majumdar at Princeton University.</i><br>Formulation and solution of equations governing the dynamic behavior of engineering systems, primarily focused on rigid-body Newtonian mechanics. Emphasis is placed on analytical rigor matched with computational validation via ODE solvers in MATLAB.                                                                                                                                                                                                          |
| Spring ‘18      | <b>Teaching Assistant, ENAE 432: Control of Aerospace Systems</b><br><i>Taught by R. Sanner at the University of Maryland.</i><br>An introduction to the feedback control of dynamic systems in the SISO setting. Topics include: Laplace transforms and transfer function techniques; frequency response and Bode diagrams; stability analysis via root locus and Nyquist techniques; performance specifications in time and frequency domains; design of compensation strategies to meet performance goals. Material includes both theoretical analysis and computational validation in MATLAB. |
| Spring ‘16      | <b>Teaching Assistant, FIRE 198: First-Year Innovation and Research Experience –Autonomous Unmanned Systems Stream</b><br><i>Taught by D. Yeo at the University of Maryland.</i><br>A hardware-based introduction to unmanned robotic systems, in which students developed micro-UAV platforms to apply to research applications, including formation flight, environmental mapping, and vision-based navigation.                                                                                                                                                                                 |

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## Refereed Conference Publications

- [CP13<sup>†</sup>] Apurva Badithela\*, David Snyder\*, Tenny Yin, Jade Sceats, George J. Pappas, Nikolai Matni, Masha Itkina, Haruki Nishimura, and Anirudha Majumdar. “AnyRank: Reliable Anytime Policy Ranking via Sequential Testing;” *Under Review*.
- [CP12<sup>†</sup>] Zachary Ravichandran, David Snyder, Alexander Robey, Hamed Hassani, Vijay Kumar, and George J. Pappas. “Contextual Safety Reasoning and Grounding for Open-World Robots;” *Under Review*.
- [CP11] Pavlos Kallinikidis\*, Fengjun Yang\*, David Snyder, Jacob H. Seidman, Nikolai Matni, Paris Perdikaris, and George J. Pappas. “Physics-Informed Learning of Feedback-Linearizing Representations;” Accepted to *Conference on Decision and Control (CDC)* (2026).
- [CP10] **David Snyder**, Apurva Badithela, Nikolai Matni, George J. Pappas, Anirudha Majumdar, Masha Itkina, and Haruki Nishimura. “Beyond Binary Success: Sample-Efficient and Statistically Rigorous Robot Policy Comparison;” Proceedings of *Robotics: Science and Systems (RSS)* (2026).
- [CP9] Apurva Badithela, David Snyder, Lihan Zha, Joseph Mikhail, Matthew O’Kelly, Anushri Dixit, and Anirudha Majumdar. “Reliable and Scalable Robot Policy Evaluation with Imperfect Simulators;” Proceedings of *IEEE International Conference on Robotics and Automation (ICRA)* (2026).
- [CP8] **David Snyder**, Asher James Hancock, Apurva Badithela, Emma Dixon, Patrick Miller, Rares Andrei Ambrus, Anirudha Majumdar, Masha Itkina, and Haruki Nishimura. “Is Your Imitation Learning Policy Better than Mine? Policy Comparison with Near-Optimal Stopping;” Proceedings of *Robotics: Science and Systems (RSS)* (2025).
- [CP7] Eric Lepowsky\*, **David Snyder\***, Alexander Glaser, and Anirudha Majumdar. “Privacy-Preserving Map-Free Exploration for Confirming the Absence of a Radioactive Source;” *IEEE International Conference on Intelligent Robots and Systems* (2024).
- [CP6] **David Snyder**, Meghan Booker, Nathaniel Simon, Wenhan Xia, Daniel Suo, Elad Hazan, and Anirudha Majumdar. “Online Learning for Obstacle Avoidance;” *7th Annual Conference on Robot Learning (CoRL)* (2023).
- [CP5] Nathaniel Simon, Allen Z. Ren, Alexander Piquè, David Snyder, Daphne Barretto, Marcus Hultmark, and Anirudha Majumdar. “FlowDrone: Wind Estimation and Gust Rejection on UAVs Using Fast-Response Hot-Wire Flow Sensors;” Proceedings of *IEEE International Conference on Robotics and Automation (ICRA)* (2023).
- [CP4] Alec Farid\*, **David Snyder\***, Allen Z. Ren, and Anirudha Majumdar. “Failure Prediction with Statistical Guarantees for Vision-Based Robot Control;” Proceedings of *Robotics: Science and Systems* (2022).
- [CP3] Udaya Ghai\*, **David Snyder\***, Anirudha Majumdar, and Elad Hazan. “Generating Adversarial Disturbances for Controller Verification;” Proceedings of the *3rd Conference on Learning for Dynamics and Control* (2021), PMLR 144:1192-1204.  
**Oral Presentation (< 10%).**
- [CP2] Justin Lidard, Debdipta Goswami, David Snyder, Girguis Sedky, Anya Jones, and Derek A. Paley. “Output Feedback Control for Lift Maximization of a Pitching Airfoil”, AIAA Guidance, Navigation, and Control Conference, AIAA SciTech Forum, (AIAA 2020-1836).

- [CP1] Nathan Lauer, Derrick W. Yeo, David A. Snyder, and Derek A. Paley. "Tip-Vortex Localization for Cross-Stream Position Control of a Multi-Hole Probe Relative to a Stationary Wing in a Free-Jet Wind Tunnel", AIAA Guidance, Navigation, and Control Conference, AIAA SciTech Forum, (AIAA 2017-1484).

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## Journal Publications

- [JP3<sup>†</sup>] Lihan Zha, Apurva Badithela, Michael Zhang, Justin Lidard, Jeremy Bao, Emily Zhou, David Snyder, Allen Z. Ren, Dhruv Shah, and Anirudha Majumdar. "Guiding Data Collection via Factored Scaling Curves;" *Under Review*.
- [JP2] Nathaniel Simon, Alexander Pique, David Snyder, Kyle Ikuma, Anirudha Majumdar, and Marcus Hultmark. "Fast-Response Hot-Wire Flow Sensors for Wind and Gust Estimation on UAVs", *Measurement Science and Technology* 34 (2023) 025109.
- [JP1] Justin M. Lidard, Debdipta Goswami, David Snyder, Girguis Sedky, Anya R. Jones, and Derek A. Paley. "Feedback Control and Parameter Estimation for Lift Maximization of a Pitching Airfoil", *Journal of Guidance, Control, and Dynamics* (2021) 44:3, 587-594.

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## Workshop Submissions

- [WS3] David Snyder, Apurva Badithela, Nikolai Matni, George J Pappas, Anirudha Majumdar, Masha Itkina, and Haruki Nishimura. "Sample-Efficient and Statistically Rigorous Policy Comparison Using General Robot Performance Measures;" in *Workshop on Manipulation Robustness: Towards Human-Level Robustness under Real-World Challenges*, International Conference on Robotics and Automation (ICRA), 2026.
- [WS2] Apurva Badithela, David Snyder, Lihan Zha, Joseph Mikhail, Matthew O’Kelly, Anushri Dixit, and Anirudha Majumdar. "Reliably Augmenting Real-World Tests with Simulation for Scalable Robot Policy Evaluation;" in *2nd Workshop on Safe and Robust Robot Learning for Operation in the Real World*, Conference on Robot Learning (CoRL), 2025. **Best Workshop Paper Award.**
- [WS1] Eric Lepowsky\*, and David Snyder\*. "Realizable Deployment of Limited-Knowledge Robotic Inspectors for Nuclear Verification;" in *Workshop on Bridging the Lab-to-Real Gap: Conversations with Academia, Industry, and Government*, International Conference on Robotics and Automation (ICRA), 2023.

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## Theses

- [TH] *Nonasymptotic Methods for Guaranteed Robotic Policy Synthesis and Evaluation*  
Dept. of Mechanical and Aerospace Engineering, Princeton University  
Advisor: Anirudha Majumdar

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## Patents

- [PA1] N. Simon, A. Pique, D. Snyder, K. Ikuma, A. Majumdar, and M. Hultmark. "Omnidirectional Flow Sensor," U.S. Patent 12 553 754, Feb. 17, 2026.

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## Professional Activities

### Workshop Organization

- [WO2] *2<sup>nd</sup> Workshop on OOD Generalization in Robotics*  
At Robotics: Science and Systems (2025), Los Angeles, CA.

[WO1] *OOD Generalization in Robotics: Towards Reliable Learning-Based Autonomy*  
At Conference on Robot Learning (2023), Atlanta, GA.

## Professional Memberships

IEEE Student Member, 2023.

## Peer Review

**Conferences:** International Conference on Robotics and Automation (ICRA); Neural and Information Processing Systems (NeurIPS); Conference on Robot Learning (CoRL).

**Journals:** International Journal of Robotics Research (IJRR); IEEE Robotics and Automation Letters (RA-L).